

## For loop

for variable in iterable:

iterable: Any object you can iterate over like a list, string, tuple, range, dictionary, etc.

variable: A placeholder name that takes the value of each item in the iterable.

```
In [2]: students = ["yasir", "arafat"]

for x in students:
    print(x)
```

```
yasir
arafat
```

```
In [3]: for i in range(5):
        print(i)
```

```
0
1
2
3
4
```

```
In [4]: for x in "yasir":
        print(x)
```

```
y
a
s
i
r
```

```
In [5]: for i in range(5):
        print(i)
        else:
            print("end")
```

```
0
1
2
3
4
end
```

```
In [6]: for i in range(5):
        if i == 3:
            break
        print(i)
```

```
0
1
2
```

```
In [7]: for i in range(5):
        if i == 3:
            continue
        print(i)
```

0  
1  
2  
4

```
In [8]: students = ["yasir","arafat"]

for index,student in enumerate(students):
    print(index,student)
```

0 yasir  
1 arafat

```
In [9]: x = {
        "name" : "yasir",
        "age" : 20
    }
    for key,value in x.items():
        print(key, "=", value)
```

name = yasir  
age = 20

```
In [10]: for i in range(3):
        for j in range(3):
            print(f"i={i},j={j}")
```

i=0,j=0  
i=0,j=1  
i=0,j=2  
i=1,j=0  
i=1,j=1  
i=1,j=2  
i=2,j=0  
i=2,j=1  
i=2,j=2

```
In [11]: squ = [x**2 for x in range(5)]
        print(squ)
```

[0, 1, 4, 9, 16]

In [ ]: